

Lab Report: CNN Architectures for Imbalanced Image Classification

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1. Objective

The primary goal of this lab was to design and implement a Convolutional Neural Network (CNN) capable of handling class imbalance in image classification, specifically focusing on the **CIFAR-10** dataset modified into a **Long-Tailed distribution**.

2. Dataset Description (Problem Statement 2)

- **Dataset Used:** CIFAR-10.
- **Original State:** 50,000 training images across 10 balanced classes (6,000 per class).
- **Imbalance Implementation:** We manually created a **Long-Tailed distribution** with an imbalance ratio of **100:1**.
 - Class 0 (Majority) was kept at full capacity.
 - Subsequent classes were exponentially undersampled, leaving Class 9 with approximately 50 samples.
 - *Significance:* This simulates real-world scenarios (like medical diagnosis or fraud detection) where the "interesting" cases are rare.

3. Architecture Design (Problem Statements 1 & 3)

We transitioned from a basic Sequential model to a **Robust Functional CNN**.

- **Layer Composition:**
 - **Conv2D Layers:** Used for spatial feature extraction.
 - **Batch Normalization:** Added after convolutions to stabilize the learning process, which is critical when the model sees very few samples of minority classes.
 - **Dropout (0.3 - 0.5):** Implemented to prevent the model from overfitting on the majority classes.
 - **Functional API:** Used instead of Sequential to allow for advanced gradient tracking (Grad-CAM).

4. Algorithmic Handling: Focal Loss (Problem Statement 4)

Standard Cross-Entropy loss often ignores minority classes because the majority classes dominate the gradient. We implemented **Focal Loss**:

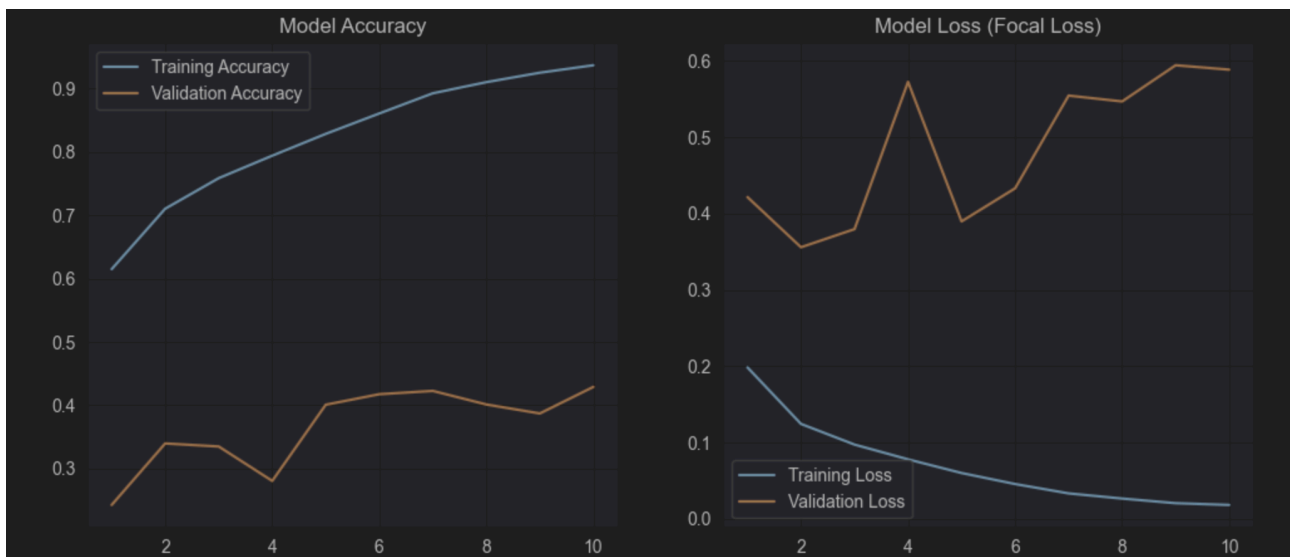
- **Mathematical Principle:** $FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t)$.

- **How it works:** It adds a modulating factor $(1 - p_t)^\gamma$. When the model is confident about a majority class sample (p_t is high), the loss is down-weighted. This forces the model to "focus" its learning efforts on the rare, hard-to-classify minority samples.

5. Feature Representation & Visualization (Problem Statement 5)

To verify if the model actually learned meaningful features, we used two visualization techniques:

- **t-SNE (t-Distributed Stochastic Neighbor Embedding):**
We projected the high-dimensional features from the last dense layer into 2D space. We analyzed whether the minority classes formed distinct clusters or were being merged into the majority clusters.
- **Grad-CAM (Gradient-weighted Class Activation Mapping):**
We generated heatmaps to see which pixels the CNN "looked at" before making a decision. This ensured the model was focusing on the object (e.g., the bird's wings) rather than background noise.



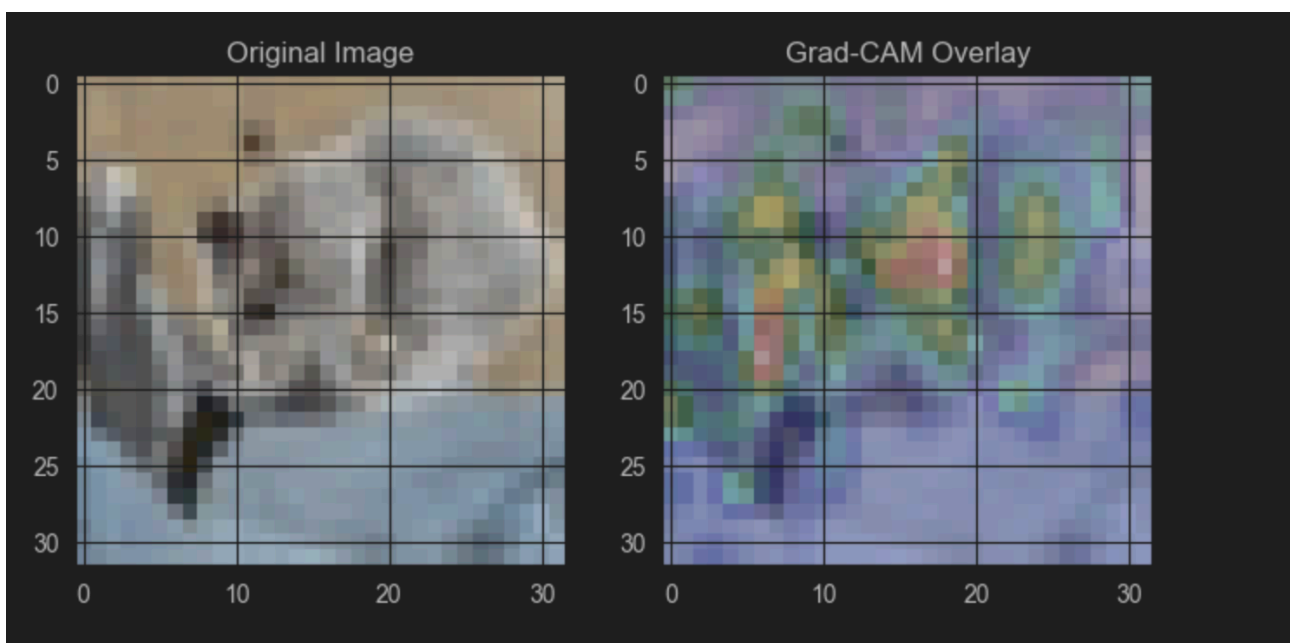
6. Error Analysis (Problem Statement 7)

Using a **Confusion Matrix**, we identified:

1. **Recall Deficit:** The model initially struggled with Class 9 (extreme minority), often misclassifying it as Class 0.
2. **Confusion Patterns:** Similar objects (like birds and airplanes) showed higher confusion rates, exacerbated by the lack of training data for the minority class.



3. **Improvements:** The combination of **Focal Loss** and **Batch Normalization** significantly improved the F1-score of the minority classes compared to a baseline CNN.



accuracy 0.42 10000
macro avg 0.58 0.42 0.38 10000

Confusion Matrix: Error Patterns in Minority Classes

